

PRODUCT REQUIREMENTS DOCUMENT

PM Agent MVP

From customer signal to engineering-ready work

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Product stage: Deployable MVP / productionization required

Purpose of this PRD

Align the team on the MVP user problem, product scope, functional requirements, production gaps, release criteria, and measurable outcomes. This document describes the current repository honestly: the experience is demo-ready and deployable, while persistence, real authentication, AI grounding, and several external integrations still require production implementation.

Core product promise

PM Agent converts scattered support signals into explainable, reviewable, engineering-ready tickets without removing human control.

1. Executive Summary

PM Agent is a product operations workspace for Product Managers and Customer Support teams. It ingests customer requests, enriches them with product and code context, drafts structured specifications, and routes approved work toward engineering delivery. The product is designed to reduce manual ticket rewriting, shorten triage time, improve consistency, and preserve human accountability at every consequential decision.

MVP release definition

A team can experience the full product journey in a deployed environment: sign in or onboard, connect representative sources, ask product questions, review AI-drafted tickets in Pipeline, apply ticket decisions, inspect insights, configure rules, and exercise available API routes. Production release requires the hardening work listed in Section 12.

1.1 Problem statement

- Customer issues arrive across disconnected channels and are rewritten manually before engineering can act.
- Support agents often interrupt PMs to decide whether a report is expected behavior, a bug, or a feature request.
- Ticket quality varies: context, reproduction detail, scope, code references, and acceptance criteria are frequently incomplete.
- Existing queues show volume but do not consistently explain priority, confidence, or the next best action.
- Automation without an audit trail creates trust and accountability concerns.

1.2 Product vision

Create one coherent workspace where teams can move from signal to decision to delivery, with grounded context, explicit guardrails, and a visible human approval step.

1.3 Product principles

Principle	Product implication
Human control	No consequential escalation or engineering handoff occurs without an explicit user action.
Explainability	Drafts expose classification, confidence, rationale, evidence, and related code references.
Action over reporting	Every primary screen should answer what the user should do next.
Traceability	Decisions, edits, overrides, and automation outcomes must be attributable and reviewable.
Focused AI	Ask PM is scoped to connected product context and must decline unsupported general conversation.

2. Users and Jobs to Be Done

2.1 Primary users

User	Primary job	Key need
Product Manager	Turn customer evidence into prioritized, reviewable engineering work.	Fast decisions with context, confidence, and control.
Customer Support Agent	Resolve product questions or escalate genuine issues without interrupting PMs.	Grounded answers and a safe, confirmed ticket path.
Engineering / Dev Agent	Receive a clear implementation-ready specification.	Consistent scope, acceptance criteria, code references, and source context.
Workspace Admin	Configure sources, repositories, rules, team access, and operational settings.	Reliable setup, RBAC, auditability, and integration health.

2.2 Core jobs

- When a customer report arrives, classify and scope it consistently so the right work enters the queue.
- When a support agent has a product question, answer from approved code and documentation or propose a confirmed escalation.
- When a PM reviews a draft, show enough evidence to accept, edit, reject, ignore, or investigate without switching tools.
- When work is approved, package it for the selected project tool or development agent with complete context.
- When performance changes, surface actionable insights and the underlying tickets rather than charts alone.

2.3 Roles and decision rights

Capability	Owner/Admin	User
Review and edit ticket drafts	Allowed	Allowed
Accept, reject, or ignore tickets	Allowed	Allowed
Configure integrations and automation	Allowed	Restricted by policy
Invite or remove members	Allowed	Not allowed
View audit history	Allowed	Read access as configured

3. Scope and Current Implementation

The current repository is a Next.js 16 and React 19 MVP designed for Vercel. It includes production-style application routes and API handlers, but several core services still use mock data, browser persistence, or process-local server memory. The distinction below is mandatory for planning and stakeholder communication.

Area	Current MVP state	Production target
Authentication	Sign-in, sign-up, reset, verification, and redirect UI; workspace state simulates session behavior.	Managed identity provider, secure sessions, email verification, password reset, and server-side authorization.
Onboarding	Guided source, repository, output, product-doc, and indexing experience using representative catalogs and delays.	Real OAuth/API connections, durable credentials, indexing jobs, retries, and progress events.
Ask PM	Global and ticket-context chat surfaces with scripted product responses and escalation behavior.	Grounded retrieval over authorized repositories/docs, citations, safety boundaries, and evaluation monitoring.
Pipeline	Review workspace, filters, clusters, ticket detail, reasoning, editing, decisions, and delivery states.	Durable collaborative queue with concurrency control, audit history, and real downstream delivery.
Ingestion APIs	Freshdesk and generic webhook routes normalize and create draft tickets in process-local memory.	Authenticated, idempotent ingestion backed by a database and asynchronous processing.
Delivery / dispatch	Payload generation, outbound dev-agent webhook, simulated external ticket URLs, and logged notification payloads.	Real Jira/Linear/etc. adapters, email provider, retries, delivery status, and reconciliation.
Analytics / settings	Functional UI over seeded and locally persisted state.	Event-backed metrics, tenant-scoped configuration, durable audit logs, and policy enforcement.

3.1 In scope for the MVP experience

- Authentication and onboarding user journeys.
- Ask PM global chat and ticket-context chat entry points.
- Ticket ingestion, classification, scope, confidence, code references, and human review actions.
- Canonical Pipeline workspace with ticket and cluster deep links.
- Connections, repositories, product documents, automation presets/rules, team, theme, and settings surfaces.
- Insights, activity, alerts, delivery stages, audit concepts, and downstream payload generation.
- Vercel-compatible build and runtime API routes.

3.2 Explicitly out of scope for this release

- Autonomous code modification or pull-request creation by Ask PM.
- Silent ticket creation without user confirmation.
- General-purpose AI conversation outside connected product context.
- Claims of production-grade durability, security, or multi-tenant isolation before Section 12 gates are complete.

4. End-to-End Product Flow

1. A user signs in or creates a workspace and completes onboarding.
2. The workspace connects a customer-ticket source, one or more repositories, an output tool, and optional product documents.
3. Inbound requests arrive through a connector or webhook and are normalized into the common ticket model.
4. PM Agent classifies the request, estimates scope, identifies related code context, and creates a draft specification with confidence and reasoning.
5. The draft appears in Pipeline, where a PM or authorized user reviews the source conversation, evidence, draft, and acceptance criteria.
6. The reviewer accepts for development, accepts as non-technical, edits, rejects, ignores, or opens Ask PM with ticket context.
7. Accepted engineering work is packaged for a project tool or dev agent; delivery status progresses from accepted to shipped.
8. Events feed Insights, audit history, operational alerts, and improvement metrics.

Guardrail

AI prepares and recommends; a human confirms. The product must retain source context, the generated draft, the final approved version, and the actor responsible for each decision.

4.1 Primary navigation

Destination	Purpose
Ask PM	Ask grounded product questions or investigate a selected ticket.
Pipeline	Review, prioritize, edit, and route tickets and clusters.
Connections	Manage sources, repositories, outputs, and product context.
Automation	Configure presets and readable if/then rules.
Settings	Manage landing preference, notifications, workspace, profile, and product behavior.
Insights / Team	Inspect outcomes, governance, roles, and accountability where enabled.

5. Functional Requirements

5.1 Authentication and onboarding

ID	Requirement	Priority
FR-001	Users must be able to sign up, sign in, sign out, reset a password, and verify an email through a real identity service.	Must
FR-002	New workspaces must complete a guided setup for ticket sources, repositories, output tools, and optional product documentation.	Must
FR-003	Connection tests and indexing must report progress, completion, recoverable failures, and retry actions.	Must
FR-004	The workspace must preserve onboarding completion and redirect authenticated users to their configured landing page.	Must

5.2 Ticket ingestion and enrichment

ID	Requirement	Priority
FR-005	The system must accept authenticated Freshdesk and generic webhook payloads and normalize them into one ticket schema.	Must
FR-006	Duplicate source events must be detected using tenant, provider, and external ticket identifiers.	Must
FR-007	Each draft must include classification, scope, description, suggested approach, acceptance criteria, confidence, rationale, and available code references.	Must
FR-008	Original message, customer, source, attachments, notes, and conversation history must remain linked to the generated draft.	Must
FR-009	Low-confidence or insufficient-context tickets must be flagged for manual review rather than silently completed.	Must

5.3 Ask PM

ID	Requirement	Priority
FR-010	Ask PM must support a global conversation and a ticket-context conversation that carries the selected ticket automatically.	Must
FR-011	Answers must be grounded only in authorized repositories, product documents, and ticket history, with source citations where available.	Must
FR-012	Simple questions may be answered directly; material issues must produce a proposed ticket summary and wait for confirmation.	Must
FR-013	Ask PM must not edit code, create pull requests, or operate as an unrestricted general assistant.	Must
FR-014	A ticket created from chat must retain the originating conversation and be labeled as chat-originated.	Should

5.4 Pipeline review and decisions

ID	Requirement	Priority
FR-015	Pipeline must show pending tickets and clusters with filters for classification, scope, priority, confidence, age, and search.	Must
FR-016	Selecting a ticket must show source context, editable draft fields, code references, confidence, and a reasoning trace without leaving the workspace.	Must
FR-017	Authorized users must be able to accept for development, accept as non-technical, edit, reject, ignore, and undo eligible decisions.	Must
FR-018	Cluster actions must support safe bulk review while retaining per-ticket history.	Should
FR-019	Deep links must open the intended ticket or cluster and preserve supported query filters.	Must
FR-020	Every decision and material edit must create an audit event with actor, time, object, and before/after context.	Must

5.5 Delivery, connections, automation, and insights

ID	Requirement	Priority
FR-021	Approved engineering work must generate a deterministic payload for the configured output tool or dev agent.	Must
FR-022	Outbound delivery must expose queued, delivered, failed, retrying, and reconciled states; simulated URLs are not acceptable in production.	Must
FR-023	Connections must display provider health, credential status, last successful sync, indexing state, and actionable errors.	Must
FR-024	Automation must offer conservative, balanced, and aggressive presets plus readable if/then rules with preview before activation.	Should
FR-025	Insights must connect metrics and recommendations to the underlying tickets and support delivery, quality, and chat-origin performance views.	Should
FR-026	Settings must support theme, default landing page, notification preferences, workspace preferences, and tenant-scoped policy defaults.	Should

6. Data, API, and Integration Requirements

6.1 Core data entities

Entity	Minimum durable data
Workspace	Tenant identity, plan, settings, lifecycle state, and retention policy.
User / membership	Identity, workspace membership, role, status, and last access.
Integration	Provider, type, encrypted credential reference, health, sync cursor, and error state.
Ticket	Source identity, customer, conversation, classification, scope, draft, final decision, and timestamps.
Evidence	Code/document reference, citation metadata, indexing version, and relevance score.
Delivery	Destination, payload version, external ID/URL, state, attempts, and reconciliation result.
Audit event	Actor, action, target, tenant, timestamp, correlation ID, and before/after summary.

6.2 Existing API surface

Endpoint group	Purpose	Production note
/api/webhooks/*	Ingest Freshdesk or generic source events.	Require authentication, signature validation, replay protection, idempotency, and rate limits.
/api/draft-tickets/*	List/create drafts and record review decisions.	Require tenant scoping, authorization, validation, persistence, and optimistic concurrency.
/api/product-docs	List and register product documents.	Add object storage, malware scanning, parsing jobs, versioning, and deletion policy.
/api/deliver	Build downstream issue payload.	Replace simulated external IDs/URLs with provider adapters and reconciliation.
/api/dispatch	Send an approved ticket to a configured development-agent webhook.	Use allow-listed destinations, signed requests, retry queue, timeout policy, and secret rotation.
/api/notify-dev	Prepare developer notification content.	Connect a real email/notification provider and delivery status tracking.

6.3 Environment and secrets

- FRESHDESK_WEBHOOK_SECRET must be configured in deployed environments; an unset secret must not leave a production webhook open.
- FRESHDESK_API_KEY status alone is not a complete integration; credentials, domain/account identity, and actual API calls must be implemented securely.
- Provider credentials and webhook destinations must never be returned to the browser or stored in localStorage.
- Secrets must use Vercel environment configuration or an approved secret manager, with separate values by environment.

7. Non-Functional Requirements

Category	Requirement / target
Security	Tenant isolation, server-side authorization, encrypted credentials, signed webhooks, least privilege, secure headers, abuse controls, and dependency scanning.
Reliability	No accepted request is lost; ingestion and delivery are idempotent; failed jobs are retryable; durable backups and recovery procedures are tested.
Performance	Primary authenticated pages should become interactive within 3 seconds at p75 on a normal broadband connection; common API reads should respond within 500 ms at p95 excluding external providers.
Observability	Structured logs, request/correlation IDs, job metrics, integration health, error tracking, and alert thresholds must exist before production launch.
Accessibility	Keyboard-operable workflows, visible focus, semantic controls, labeled forms, readable contrast, and WCAG 2.1 AA as the launch target.
Privacy	Document data handling, retention, deletion, sub-processors, and code-indexing boundaries; avoid storing source content beyond approved policy.
Compatibility	Support current stable desktop Chrome, Edge, Firefox, and Safari; responsive behavior for review and chat on small screens.
Maintainability	Typed contracts, validation at boundaries, automated tests for critical flows, migration discipline, and environment-parity checks.

7.1 Audit and accountability

- Store who performed each review action and whether it was manual or automated.
- Preserve the generated draft and the final approved content.
- Record rule changes, integration changes, role changes, dispatch attempts, and delivery outcomes.
- Make audit records append-only for normal users and exportable for administrators.

8. Success Metrics

The following are proposed targets for a controlled pilot. Baselines must be measured before interpreting improvement.

Outcome	Metric	Pilot target
Faster triage	Median time from ingestion to first human decision	50% lower than baseline
Useful drafts	Accepted or edit-and-accepted drafts / reviewed drafts	>= 70%
Lower review effort	Median active review time per ticket	<= 2 minutes
Grounded answers	Ask PM answers rated useful without escalation	>= 75%
Safe escalation	Chat-created tickets with recorded user confirmation	100%
Reliable ingestion	Valid source events processed successfully	>= 99%
Delivery quality	Approved tickets delivered without manual payload repair	>= 95%
Trust	Low-confidence drafts routed to manual review	100%

8.1 Instrumentation events

- ticket_ingested, ticket_draft_created, ticket_review_opened, ticket_edited, ticket_decided
- chat_question_submitted, chat_answer_completed, chat_escalation_proposed, chat_ticket_confirmed
- delivery_started, delivery_succeeded, delivery_failed, delivery_retried
- integration_connected, integration_tested, integration_failed, indexing_started, indexing_completed
- automation_previewed, automation_enabled, automation_triggered, automation_overridden

9. MVP Acceptance Criteria

9.1 Demo acceptance

- A clean installation completes the production build and starts successfully in a Vercel-compatible environment.
- A new user can traverse authentication, onboarding, and the configured landing route without broken navigation.
- A valid test webhook creates or identifies a draft ticket and the item is visible in the review workflow.
- A reviewer can open a deep-linked ticket, inspect evidence, edit the draft, make each supported decision, and observe the resulting state.
- Ask PM supports both global and ticket context and never performs an unconfirmed escalation.
- Connections, Automation, Settings, Insights, and Team surfaces render with coherent empty, loading, success, and error behavior.
- The primary light and dark themes remain readable and consistent across the core routes.

9.2 Quality acceptance

- No build-blocking TypeScript errors, high-severity dependency vulnerabilities, or leaked secrets.
- Critical paths have automated tests for authorization, ingestion idempotency, review decisions, and delivery failure handling.
- No P0/P1 defects remain open; known P2 defects have documented owners and release decisions.
- Accessibility review covers keyboard navigation, forms, dialogs, contrast, focus, and screen-reader labels.
- Analytics events and error monitoring are verified in the target environment.

10. Risks and Mitigations

Risk	Impact	Mitigation
Prototype state presented as production-ready	Stakeholder trust, data loss, or security exposure.	Use the release gates in Section 12 and label demo environments clearly.
In-memory or browser-only state	Tickets, audits, and settings can disappear or diverge.	Move tenant data to a transactional database with migrations and backups.
Ungrounded AI response	Incorrect product guidance or poor engineering work.	Use retrieval citations, confidence thresholds, evaluations, and human confirmation.
Weak webhook security	Unauthorized ticket creation or replay attacks.	Require signatures/secrets, timestamps, nonce/replay controls, rate limits, and idempotency.
External provider failure	Stalled ingestion or delivery.	Queue work, retry safely, show status, alert owners, and reconcile provider state.
Over-automation	Incorrect tickets are accepted or dispatched.	Conservative defaults, previews, bounded rules, approval thresholds, undo, and audit logs.
Sensitive code/document exposure	Privacy and intellectual-property risk.	Tenant isolation, access-scoped indexing, encryption, retention controls, and redaction.

11. Delivery Plan

Phase	Outcome	Exit criteria
Phase 0 - Demo stabilization	Deployable team demo with coherent core routes and representative data.	Build green, smoke tests pass, no broken primary navigation, demo script approved.
Phase 1 - Durable foundation	Real auth, tenant database, authorization, audit, secrets, and job infrastructure.	Data survives deploys; cross-tenant tests pass; webhook and session security reviewed.
Phase 2 - Real integrations	Freshdesk ingestion, repository indexing, product-doc processing, and Jira/Linear delivery.	Sandbox end-to-end flows pass with retries, idempotency, and reconciliation.
Phase 3 - Grounded intelligence	Evaluated classification, scoped retrieval, cited Ask PM answers, and confidence policy.	Offline evaluation thresholds and pilot quality metrics are met.
Phase 4 - Controlled pilot	Limited customer teams operate real workflows with support and monitoring.	Pilot acceptance, security review, runbooks, backups, and go/no-go decision complete.

11.1 Recommended implementation order

1. Identity, workspace tenancy, RBAC, and durable database schema.
2. Secure/idempotent ingestion plus background processing and audit events.
3. Repository and product-document indexing with tenant-scoped retrieval.
4. Grounded classification, drafting, Ask PM answers, confidence policy, and evaluations.
5. Real output adapters, notifications, delivery reconciliation, and operational dashboards.
6. Pilot hardening: tests, accessibility, security, privacy, performance, runbooks, and support readiness.

12. Production Release Gates

Decision rule

The deployed MVP may be shared as a product demonstration. It must not be described as production-ready until every Must gate below is verified in the production environment.

Gate	Required evidence	Priority
Persistent data	Database migrations, tenant-scoped schema, backups, restore test, and removal of process-local state for business records.	Must
Real authentication	Identity provider, secure cookies/sessions, verification/reset flow, RBAC, and server-side access tests.	Must
Webhook security	Configured secrets/signatures, timestamp validation, replay protection, idempotency, rate limits, and monitoring.	Must
AI grounding	Authorized index, citations, evaluation set, confidence policy, refusal behavior, and prompt/data isolation tests.	Must
Real delivery	At least one supported source and one supported output operate end to end without simulated URLs or logged-only notifications.	Must
Operational readiness	Error tracking, dashboards, alerting, runbooks, owner rotation, incident process, and dependency/security scan.	Must
Quality	Critical automated tests, accessibility review, performance baseline, cross-browser smoke tests, and no open P0/P1 issues.	Must
Privacy/legal	Data map, retention/deletion behavior, customer disclosures, provider agreements, and code/document handling approval.	Must

13. Open Decisions

Decision	Why it matters	Recommended owner
Initial pilot persona: PM-led or Support-led?	Changes onboarding, default landing, metrics, and training emphasis.	Product
First production source and output providers	Determines integration scope and pilot feasibility.	Product + Engineering
Repository indexing boundary and retention	Defines privacy, cost, grounding quality, and customer approval needs.	Security + Engineering
Automation permission model	Controls who can enable rules and which decisions may run automatically.	Product + Security
Confidence thresholds by classification	Affects review volume, quality, and risk.	Product + Data/AI
Multi-workspace membership behavior	Impacts tenancy, navigation, notifications, and billing.	Product + Platform

13.1 Definition of done for team approval

- Product confirms the users, core promise, MVP scope, and success metrics.
- Engineering confirms the current-state assessment, production gaps, architecture direction, and sequencing.
- Design confirms the end-to-end journey, required states, accessibility target, and primary navigation.
- Security/privacy confirms the minimum controls for webhooks, credentials, code/doc indexing, retention, and audit.
- The team assigns owners and target dates to production release gates and open decisions.

Appendix A - Glossary

Term	Definition
Ask PM	The grounded conversational surface for product questions and ticket investigation.
Draft ticket	An AI-prepared specification awaiting a human decision.
Pipeline	The canonical workspace for reviewing and tracking ticket work.
Classification	The product category assigned to a request, such as bug, feature request, question, or churn signal.
Scope	A relative implementation estimate: S, M, or L.
Reasoning trace	The evidence and signals displayed to explain classification, confidence, and scope.
Code reference	A linked repository file/function/line/snippet associated with a ticket.
Non-technical acceptance	Resolution of a valid ticket without engineering delivery.
Dispatch	Sending an approved specification to a development agent webhook.
Delivery	Publishing approved work to the configured project or issue-tracking tool.

Appendix B - Source Basis

This PRD was prepared from the current PM Agent repository, its implemented routes and domain model, the PM Agent 2.0 redesign proposal, the PM Agent 2.1 addendum, and the verified July 2026 deployment build. Where repository behavior is simulated or prototype-only, the document labels the production requirement separately.